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Lab #1

METER OPERATION & DESIGN (ANALOG)

Introduction

Throughout this lab, ohms law (particularly the resistance of the meterO), properties of parallel and series circuits are tools of manipulation to maintain key conditions for the meter. By adding resistors, we can control the new R_M (resistance of the meter) to not exceed the I_{FS} and to control the voltage going into the meter. By changing the scaling resistor in *series*, we can change the voltage into the meter. To control the current in the circuit, we can add a resistor in *parallel.* Most critically, the full scale current of the meter (I_{FS}) cannot be exceeded. In this lab, I_{FS} was $50\mu A$. The full scale current should be as small as possible, as it is the ultimate sensitivity and factor of how small one can measure. Analog meters are less applicable and used compared to digital meters (especially DMMs)- however, digital multimeters use many of the same concepts with a significant conversion from current to voltage based measurements. Additionally, there are some cases where analogue meters are useful and visually easier.

Objectives

The purpose of this lab work is to:

1. Review the operation of a standard Analog meter movement through measurements, in particular the relationship between IFS (Full scale meter current), RM and VM

2. Learn the application and use of a precision decade standard (on the EICO 1171 use the bottom terminals not the top/comparator terminals) in the determination of IFS and RM.

3. Review the ranging/scaling operations of an Analog meter for voltage measurements in terms of adding series/multiplier resistors and current measurements in terms of adding shunt resistors.

4. Review and understand a basic meter schematic for a standard Analog meter (Simpson 260 type). This might be done in class or as part of a homework problem

Parts:

- DC voltage supply
- 2x Decade resistor box
- Multimeter
- Analog meter with a current of 50μA

Part 1: Determination of an R_m value of an Analog Meter

The resistance of an analog meter movement (R_M) is not a fixed value across all meters, unlike digital meters. For this reason, R_M must always be measured before use. To determine the meter resistance, it is necessary to know the full-scale current (I_{FS} = 50μA). To perform the measurement, a circuit is built using a DC power supply, the analog meter, and resistance boxes (decade resistors, using the bottom terminals). The variable resistance is adjusted until the meter reaches its full-scale reading, and the corresponding voltage across the meter circuit is measured. From these values R_M can then be calculated. Recall that the equipment used is the DC power supply, analog meter (full scale 50uA), and variable resistor (decade resistance box). First, the power supply, meter and variable resistor was connected in series. Then, the resistance was gradually adjusted until the meter needle reached a full scale reading of 50 microamps. The supply voltage across the series circuit was measured.

Full scale current: I_{FS} = 50μA

Measured voltage: $V = 1.0V$

Using Ohm's law:

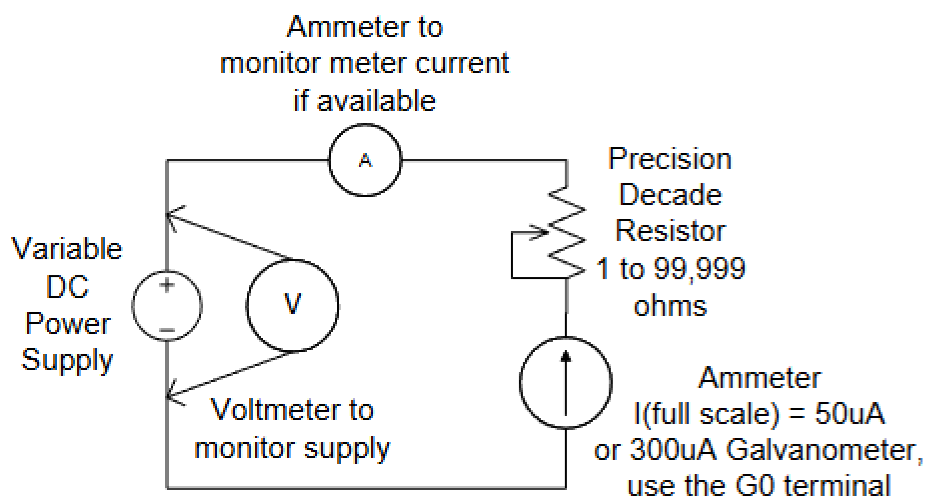
$R_{total} = \frac{V}{I_{Fs}} = \frac{1.0V}{50*10^{-6}A} = 20k\Omega$

The total resistance includes both the series resistor and the internal resistance of the meter, so $R_{total} = R_{meter} + R_{series}$. More information can be found in part 1c.

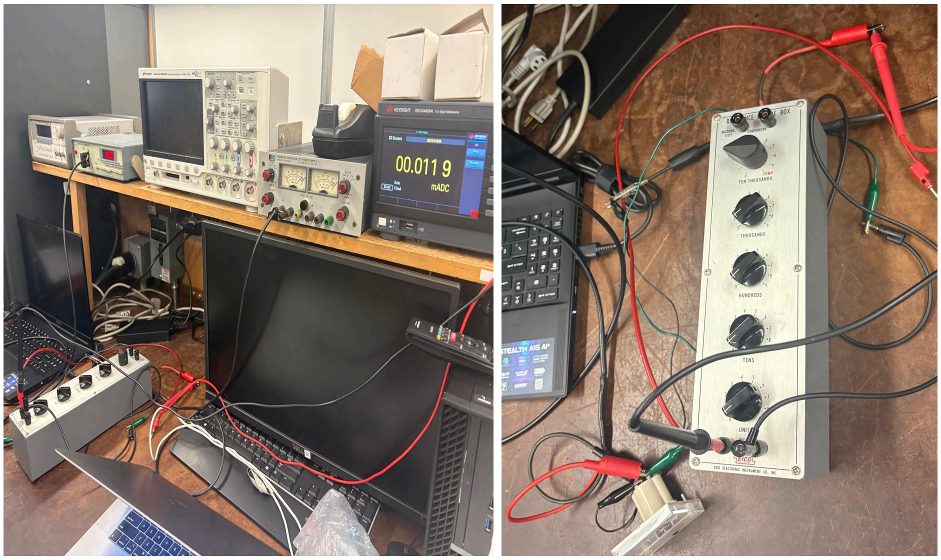
1a.

Setup a series circuit with an ammeter, variable voltage supply, precision adjustable resistor and the DUT analog meter movement. Use the bench digital meters to monitor voltages and or current as shown.

Since there was indeed a benchtop ammeter provided, the first circuit provided in the lab handout was used (image 1).



IMG 1. *Schematic of the first circuit- the voltage supply, digital ammeter, analog ammeter (50uA), and precision decade resistor in series.*



IMG 2 & 3. *The physical circuit, based off of the diagram of the first circuit (IMG 1).*

The design value of the meter is I_{FS} = 50μA. We used a voltage input of 1v (measured to be exactly $V_{in} = 1.02v$, with a digital multimeter). Note that since the circuit is in series, the current will be the same (50μA). Then , the resistance (including both the meter and external resistance) was calculated.

$$R_{total} = \frac{V_{total}}{I_{FS}} = \frac{1v}{50\mu A} = 20k\Omega.$$

The total resistance of the circuit (the meter and all other external resistance) was calculated to be $R = 20k\Omega$.

1b.

Next, we set the precision decade resistor box to max 99kohms (~100kohms) and decreased its value until the meter reads I_{FS} = 50μA, being sure to not exceed this value. Since the circuit is in series, all components carry the same current. With the supply voltage measured at $V_{in} = 1.02V$, the new total resistance of the circuit is

$R_{total} = \frac{V_{total}}{I_{FS}} = \frac{1.02v}{45*10^{-6}A} = 22.7k\Omega$. Our total resistance value of 22.7k ohms is opposed to the calculated value of 20k ohms. The error can be explained by measurement and environmental properties.

The current as read from the analog meter is visually approximate to 44mA, and the current from the digital ammeter is approximately 45mA. Note that the analog meter introduces an error of +/-2μA, which is the maximum inaccuracy of reading the analog meter. However, the digital meter was read along with the analog meter and it was ensured that the values matched throughout the lab, thus minimizing the error. This detail influences the error calculations of R_M starting in part 1c.

IMG 4. *Current read from analog meter.* **IMG 5.** *Current read from digital meter.*

Assuming the precision resistor is accurate, we were able to determine the meter resistance, RM, using the equation $R_{Total} = R_{Precision} + R_M$, or $R_M = R_{total} - R_{precision}$. Therefore, $8.8k\Omega = 22.7k\Omega - 13.9k\Omega$. The meter resistance is estimated, based on these calculations, to be $8.8k\Omega$.

However, this value for the RM is quite different then the real value of RM. This can be explained by the values of R_{Total}, $R_{Precision}$ and I_{FS}.

R_{total} has error due it being calculated based off of a I_{FS} reading of 45μA instead of the desired full scale current of 50μA. Additionally, the $R_{precision}$ value was likely recorded or adjusted inaccurately, adding further error. The resistance from measuring dirērectly across the meter in part 1c is a much more accurate data point.

1c.

R_M was confirmed by moving the voltage measurement from the power supply voltage to the meter. This voltage measurement represents V_m, and we can use the series current of I_{FS} = 50μA to calculate R_M. The voltage of the analog meter was measured to be 0.13v. The expected value of R_M is between 1-10kΩ.

$$R_M = \frac{V_M}{I_{FS}} = \frac{0.13v}{50\mu A} = 2.6k\Omega.$$

The calculated value of 2.6kΩ is between 1-10kΩ, as expected.

1d.

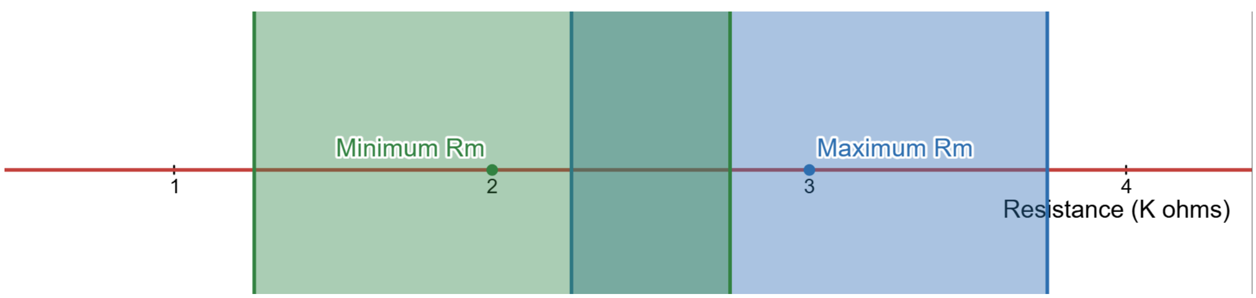
Then, we reduced the power supply voltage until the ammeter needle read 40uA. We recorded V_M and then repeated the process. It is expected that R_M should be a constant value, and close to our calculated 2.6kΩ value from part 1c.

$(I_M \pm 2)$ μA	V_M (V)	$(Calculated\ R_M \pm 0.75)$kΩ
40	0.12	3.0
30	0.08	2.7
20	0.05	2.5

10	0.02	2.0

Table 1. *The first column (I_M) represents the current as measured from the analog meter. V_M is also a measured value, and the resistance of the meter (R_M) was calculated using the formula from part 1c.*

It is expected that R_M is constant. To find the uncertainty of R_M, we can use the brute force method of $\frac{Maximum\ R_M - Minimum\ R_M}{2}$. Recall that the analog meter introduces an error of +/-2μA, which is the maximum inaccuracy of reading the analog meter. Additionally, note that the voltage was measured using a digital multimeter and so is not adjusted for error. Therefore, the minimum resistance of the meter is equivalent to $\frac{Minimum\ V_M}{I_M\ for\ minimum\ V_M + 2\mu A}$, or $\frac{0.02v}{(10\mu A + 2\mu A)}$, which is 1.7kΩ. The maximum resistance of the meter is equivalent to $\frac{Maximum\ V_M}{I_M\ for\ maximum\ V_M - 2\mu A}$, or $\frac{0.12v}{(40\mu A - 2\mu A)}$, which is 3.2kΩ. Therefore, the uncertainty for R_M is $\pm \frac{3.2k\Omega - 1.7k\Omega}{2}$, or $\pm\ 0.75k\Omega$.



IMG 6. *Graph of the maximum and minimum calculated R_M.*

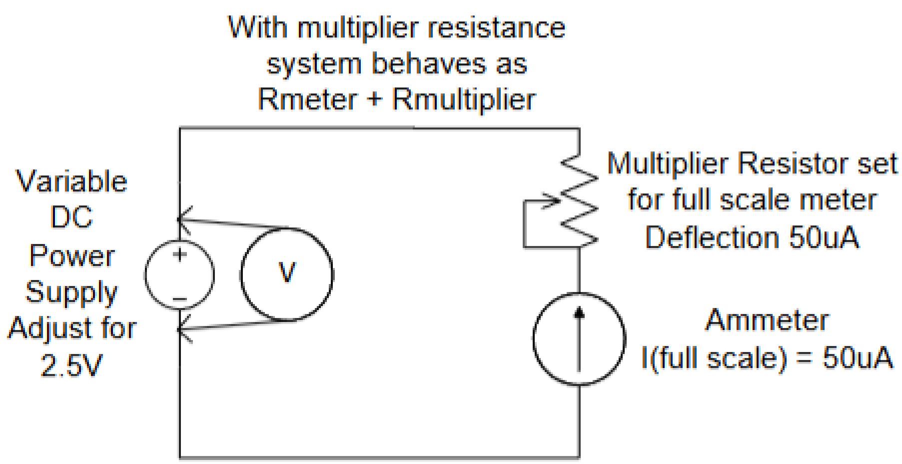
The maximum calculated R_M is shown to be within uncertainties of the minimum calculated R_M, based on image 6. R_M is therefore calculated to be 2.6kΩ $\pm\ 0.75k\Omega$.

Part 2: Determination of an R_m value of an Analog Meter

In voltmeter designs, additional ranges are created by adding resistances in series with the basic meter. When doing this, I_{FS} instead of V_M should be measured as we are operating with a series circuit. These added resistors are referred to as multiplier resistors.

2a.

The circuit shown in image 7 is built. The variable precision resistor is adjusted so that the analog meter reads full scale (I_{FS}) when the applied voltage (from the power supply) is 5V. Note that in an actual meter, the variable resistor is replaced with a single value precision resistor to create the multiplier resistor.



IMG 7. *The second circuit was set up. Our applied voltage is now 5v, and the multiplier is changed to 100k. Note that* I_{FS} *remains 50μA.*

With the measurements previously noted, we can calculate a new R_{Total}, by using the formula $R_{multiplier} = \frac{V_M}{I_{FS}} = \frac{5V}{50\mu A} = 100k\Omega$. This means that in order to have an input voltage of 5V and have I_{FS} as 50μA, we must adjust the variable precision resistor to equal $100k\Omega$.

A series circuit is still being used, so we can continue to use the formula $R_{Total} = R_{multiplier} + R_M$. Recall that R_M was calculated to be (2.6 $\pm\ 0.75)k\Omega$. Therefore, we will have a maximum total resistance of $100k\Omega\ +\ (2.6k\Omega\ +\ 0.75k\Omega\ )$, or $103\ k\Omega$. Our minimum total resistance of $100k\Omega\ +\ (2.6k\Omega\ -\ 0.75k\Omega\ )$, or $102\ k\Omega$. By using the brute force method, our error for the total resistance would be $\frac{(103-102)k\Omega}{2}$, or $\pm\ 0.5$.

Therefore, $R_{Total} = (102.6\ \pm 0.5)k\Omega$

Questions

2b.

If we wanted to have a 25V circuit, $R_{multiplier\ 25V}$ would be $497.4\ k\Omega\pm\ 1k\Omega$.

Assuming the circuit is in series,

$$R_{total} = \frac{V_{total}}{I_{FS}} = \frac{25V}{50\mu A} = 500k\Omega.$$

R_M= 2.6kΩ $\pm\ 0.75k\Omega$.

$$R_{Total} = R_{multiplier} + R_M$$

$$R_{multiplier} = R_{Total} - R_M = 500k\Omega\ -\ 2.6k\Omega = 497.4k\Omega$$

$$Max\ R_{multiplier} = R_{Total} - (R_M -\ 0.75k\Omega) = 500k\Omega\ -\ (2.6k\Omega\ -\ 0.75k\Omega) = 498k\Omega$$

$$Min\ R_{multiplier} = R_{Total} - (R_M +\ 0.75k\Omega) = 500k\Omega\ -\ (2.6k\Omega\ +\ 0.75k\Omega) = 496k\Omega$$

$$\delta\ of\ R_{multiplier} = \frac{Max\ R_{multiplier} - Min\ R_{multiplier}}{2} = \frac{498k\Omega - 496k\Omega}{2} = \pm\ 1k\Omega$$

$$R_{multiplier} = 497.4\ k\Omega \pm\ 1k\Omega$$

2c.

If we wanted to have a 250V circuit, $R_{multiplier\ 25V}$ would be $4.9M\Omega \pm\ 1k\Omega$.

Assuming the circuit is in series,

$$R_{total} = \frac{V_{total}}{I_{FS}} = \frac{250V}{50\mu A} = 5M\Omega.$$

R_M= 2.6kΩ $\pm\ 0.75k\Omega$.

$$R_{Total} = R_{multiplier} + R_M$$

$R_{multiplier} = R_{Total} - R_M = 5000k\Omega -$ 2.6kΩ= 4.9$M\Omega$

$$Max\ R_{multiplier} = R_{Total} - (R_M - 0.75k\Omega) = 5000k\Omega - (2.6k\Omega - 0.75k\Omega) = 4998k\Omega$$

$$Min\ R_{multiplier} = R_{Total} - (R_M + 0.75k\Omega) = 5000k\Omega - (2.6k\Omega + 0.75k\Omega) = 4996k\Omega$$

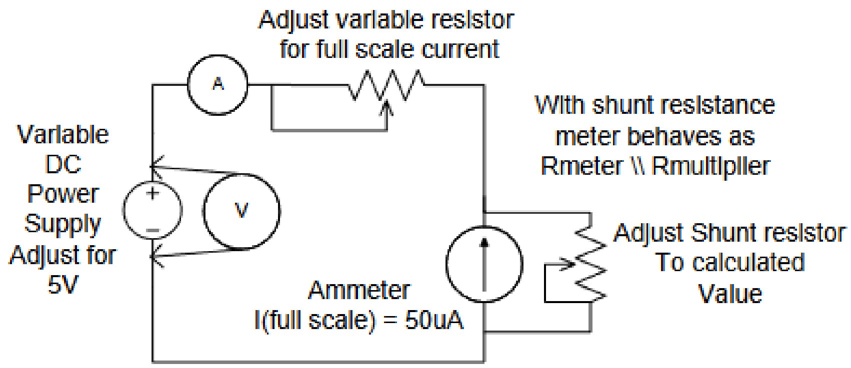
$$\delta\ of\ R_{multiplier} = \frac{Max\ R_{multiplier} - Min\ R_{multiplier}}{2} = \frac{4998k\Omega - 4996k\Omega}{2} = \pm\ 1k\Omega$$

$$R_{multiplier} = 4997.4\ k\Omega \pm\ 1k\Omega = 4.9M\Omega \pm\ 1k\Omega$$

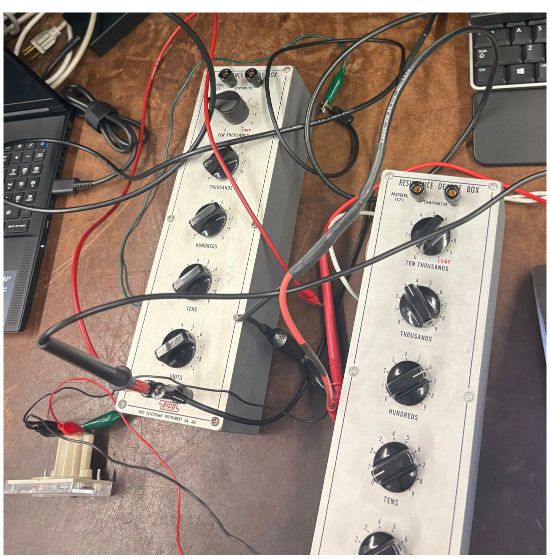
Part 3: Determination the needed resistance of the shunt value of an Analog Meter

3a.

For ammeter designs, the minimum current readings are based upon the basic meter movement I_{FS}. If I_{FS} = 50μA, for currents greater than 50μA resistors must be added in parallel. This is because the basic meter current cannot exceed 50μA without damaging the meter. Parallel resistors (which are also called shunts) serve as bypass paths for currents that are greater then I_{FS}.



IMG 8. *The schematic of the circuit used in part 3.*



IMG 9. *The physical circuit used in part 3.*

3b.

Our circuit, in image 9, was modified to have a second precision resistor in parallel with the analog meter (this is the shunt). We adjusted the shunt until we got a total supply current of 100uA. According to the diagram in image 8, the analog current should still be I_{FS} = 50μA, and the current going into the two parallel shunt resistors should also be 50μA ($I_{shunt} = I_{total} - I_{FS}$, or $50uA = 100uA - 50uA$). Therefore, we need the shunt resistors to be able to withstand a current of 50 micro amps, and we need to calculate the required added resistance using ohms law. Recall that the resistance of the meter was calculated previously in part 1c to be 2.6kohms, so the meters full scale voltage is $V_{meter\ full\ scale} = I_{FS}R_M = 50\mu A * 2.6k\Omega = 0.13V$. Since the circuit is in series, the voltage is the same, and the shunt resistance can be calculated as $R_{shunt} = \frac{V_{meter\ full\ scale}}{I_{shunt}} = \frac{0.13V}{50\mu A} = 2.6k\Omega$.After the value was calculated, we set this value on our precision resistor and confirmed that when the meter is full scale (50μA), the supply is 100μA. It was confirmed through adjusting that the shunt should be 2.6 $k\Omega$. The new shunt resistor and meter combine to essentially create a “new”” meter, ranging from (0- 100)μA.

3c.

Once we confirmed for the 100μA case, the following conditions were checked.

*When we reduce the supply to 80uA, the meter reading is at 40uA ☑

When we reduce the supply to 60uA, the meter reading is at 30uA ☑

When we reduce the supply to 40uA, the meter reading is at 20uA ☑

When we reduce the supply to 20uA, the meter reading is at 10uA ☑

*The meter readings are not exact values (ex., precisely at $40\mu A$), but close enough that it is reasonable to count these conditions to be considered true.

Part 4.

The addition of resistors can be used to control the current and voltage going to the meter. To understand how this is possible, it is critical to first fully understand aspects of the meter. The full scale current (in this case, I_{FS} = 50μA) cannot be exceeded without risking damage to the internal components of the meter, and therefore is a constant variable. We have resistance of the meter (R_M), mainly due to the force needed to move the meter. This force varies a bit from meter to meter. The DC voltage into the meter, commonly noted as V_{MFS}, is a changeable variable.

To scale the meter, we can add a resistor in *series*. By changing the scaling resistor in series, we can change the voltage into the meter. The current in the series circuit will remain the same (50μA, in this lab). To determine the resistor needed in such a case, we can use the formula $R_{multiplier} = \frac{V_{multiplier}}{I_{FS}}$, where the voltage multiplier is the minimum of the desired voltage scale. In this lab, $V_{MFS} = I_{FS} * R_m$, where I_{FS} = 50μA and $R_M = 5k\Omega$. Therefore, the acceptable voltage going into the meter is $50\mu A * 5k\Omega$, or V_{MFS} =200mV. If we want to use a 1V scale, we cannot exceed a current of $50\mu A$ and thus add a resistor in series to take up some of the voltage. The voltage quadruples ($250mv * 4$), so the resistance in the circuit should also quadruple ($5k\Omega * 4$) to maintain $50\mu A$ ($I_{FS} = \frac{1V}{20k\Omega} = 50\mu A$).

To control the current in the circuit, we can add a resistor in *parallel.* If we have 1mA going into the meter, we need to add a resistor in parallel to take $950\mu A$ in order to have the meter take in only $50\mu A$. To calculate the value needed for a resistor that can take $950\mu A$, we can do $250mV = 950\mu A * R$. Note that V_{MFS} is a constant voltage of $250mV$, because the circuit is in parallel. After solving the equation, we find that we need a resistance of 263.16Ω for a current of $950\mu A$ - of course we will not have access to a resistor of this exact value, but we can choose an approximate component.

Conclusion

Throughout this lab, ohms law and the properties of parallel and series circuits were manipulated to maintain key conditions for the meter. By adding resistors, we can control the new R_M (resistance of the meter) to not exceed the I_{FS} and to control the voltage going into the meter. By changing the scaling resistor in *series*, we can change the voltage into the meter. To control the current in the circuit, we can add a resistor in *parallel.* This lab provided clear data as well as practice supporting these concepts. The relationship between I_{FS}, R_M and V_{MFS} of a standard analog meter was thoroughly explored.